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(57) Abstract :

The System that enables users to operate home appliances using non-invasive neural signals. The system (100) includes an EEG headset (102) for capturing brain activity, a preprocessing module (104) for filtering signals, and a feature-extraction module (106) for deriving meaningful neural parameters. A classification module (108) interprets these parameters into actionable commands transmitted through a communication interface (110) to an IoT hub (112). The IoT hub (112) drives a device-control module (114) for operating appliances. Additional modules include a feedback module (116) for user confirmation, a calibration module (118) for adaptive tuning, and a security module (120) for ensuring data protection. The system (100) supports multiple neural-control paradigms (122), offering a flexible and accessible home-automation solution. FIG.1

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